Diabetes Risk Prediction Using Logistic Regression

Machine Learning Final Project:

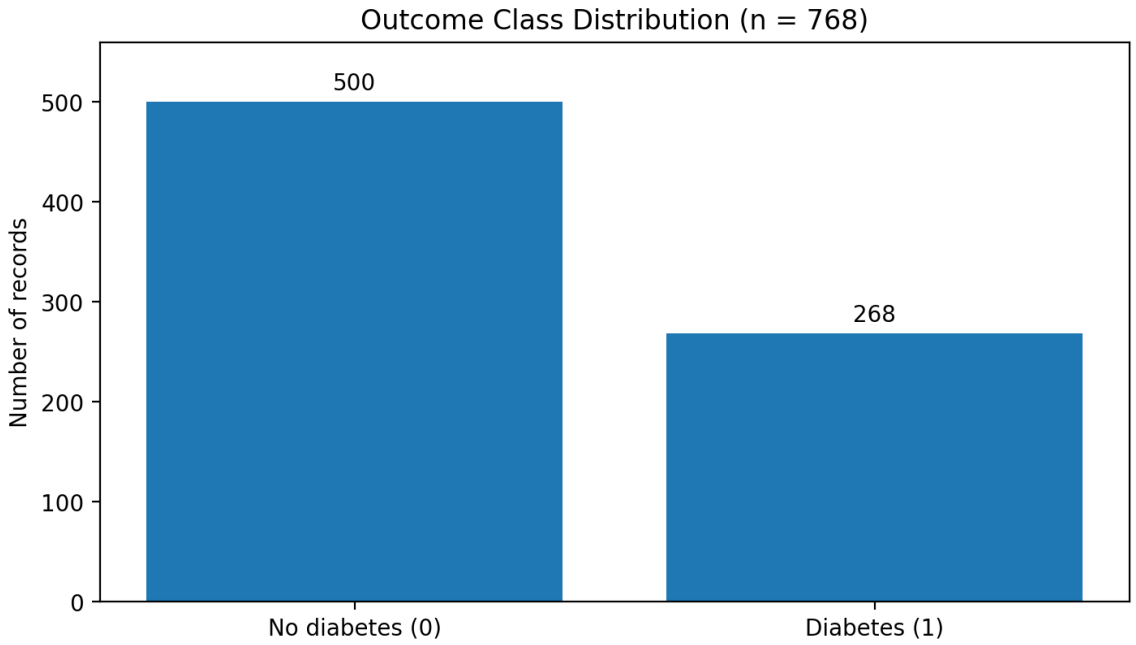
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CSBU 5420: Machine Learning for Business Analytics
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1. Project Purpose and Framework

- Prepare the Pima Indians Diabetes dataset for modeling and correct invalid measurement zeros.
- Build Logistic Regression as the primary model and compare it with five alternative classifiers.
- Evaluate performance using accuracy, AUC, precision, recall, F1, specificity, and Confusion matrices.
- Tune Logistic Regression with grid search and 10-fold cross-validation, then validate on a holdout set.
- Goal: choose the model that best balances detection of positive cases with overall holdout performance.

2. Dataset Overview

- Pima Indians Diabetes Database, 768 observation, 8 numeric predictors, binary Outcome.



Predictors

- Pregnancies
- Glucose
- BloodPressure
- SkinThickness
- Insulin
- BMI
- DiabetesPedigreeFunction
- Age

Target

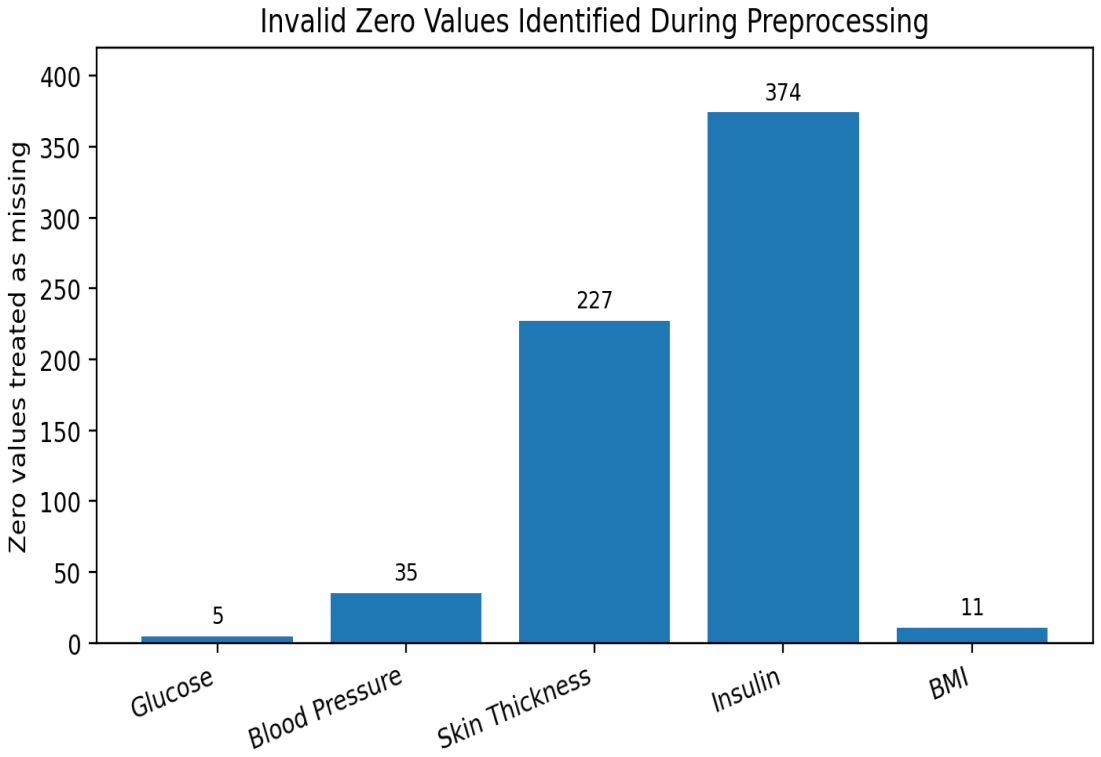
Outcome
0 = No diabetes
1 = Diabetes

34.9% of records are positive → moderate class imbalance.

- 500 Negative Cases
- 268 positive Cases

3. Data Preprocessing

Cleaning focused on biologically invalid zero values and reproducible train/test preparation.



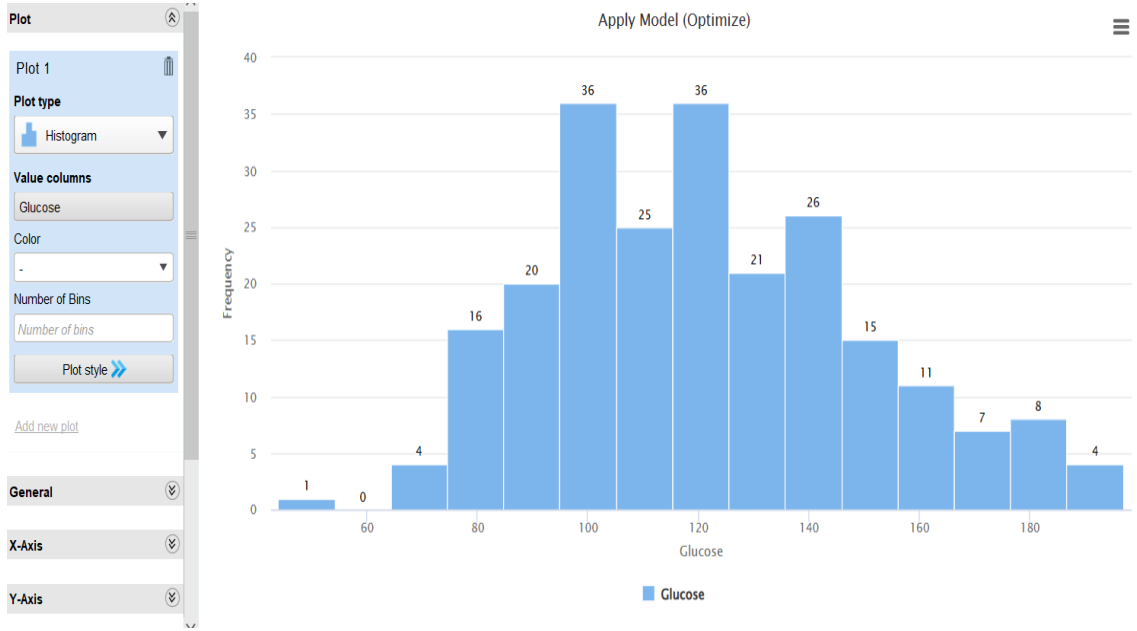
Preprocessing decisions

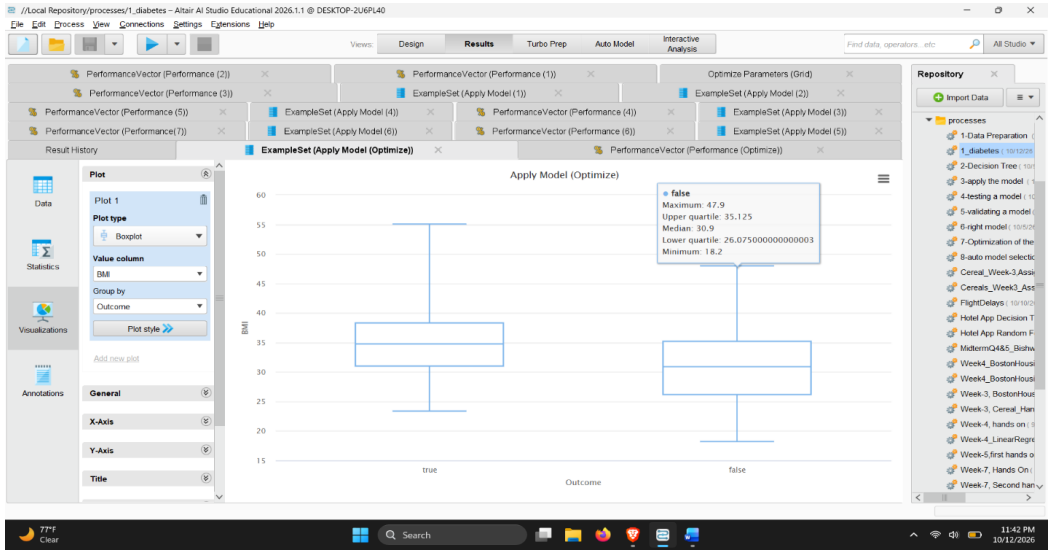
- Zeros in Glucose, BloodPressure, SkinThickness, Insulin, and BMI were treated as missing.
- Missing measurements were replaced using mean imputation.
- Outcome was defined as a binominal label (false/true).
- Stratified 70/30 split produced ~538 training and 230 testing records.
- k-NN used normalized predictors because it is distance based.

Largest data-quality issue: Insulin (374 invalid zeros) and SkinThickness (227 invalid zeros).

4. Exploratory Data Analysis

Distribution and Outcome-group difference in the held-out Exampleset





Glucose distribution

Most observations cluster in the mid-range, with fewer cases at very high glucose levels.

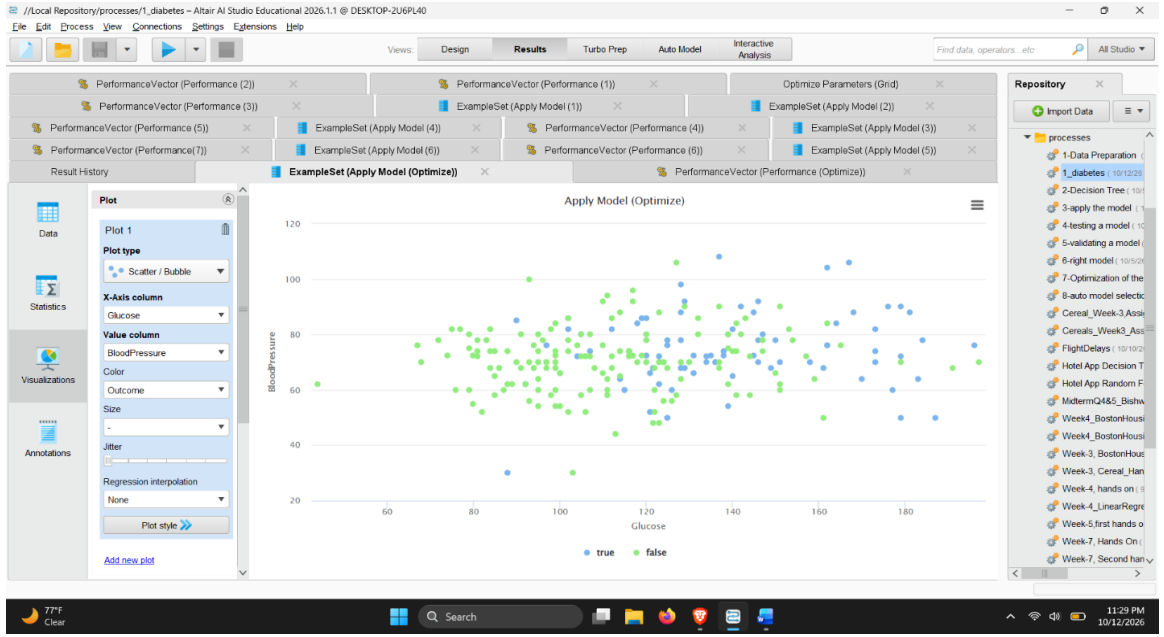
BMI by diabetes Outcome

Median BMI: ~34.75 for positive cases vs. ~30.90 for negative cases.

EDA suggests meaningful separation in Glucose and BMI between the two Outcome groups.

5. Relationship and Correlation

Bivariate scatter patterns and Pearson correlation among six selected attributes





Glucose vs. BloodPressure

Positive cases occur more often at higher glucose values; BloodPressure alone provides limited separation.

Correlation highlights

Pregnancies ↔ Age: r = 0.528

BloodPressure ↔ Age: r = 0.333

Glucose ↔ Age: r = 0.202

Most pairwise correlations are weak, so severe multicollinearity is not evident among the six displayed predictors.

6. Models Evaluated

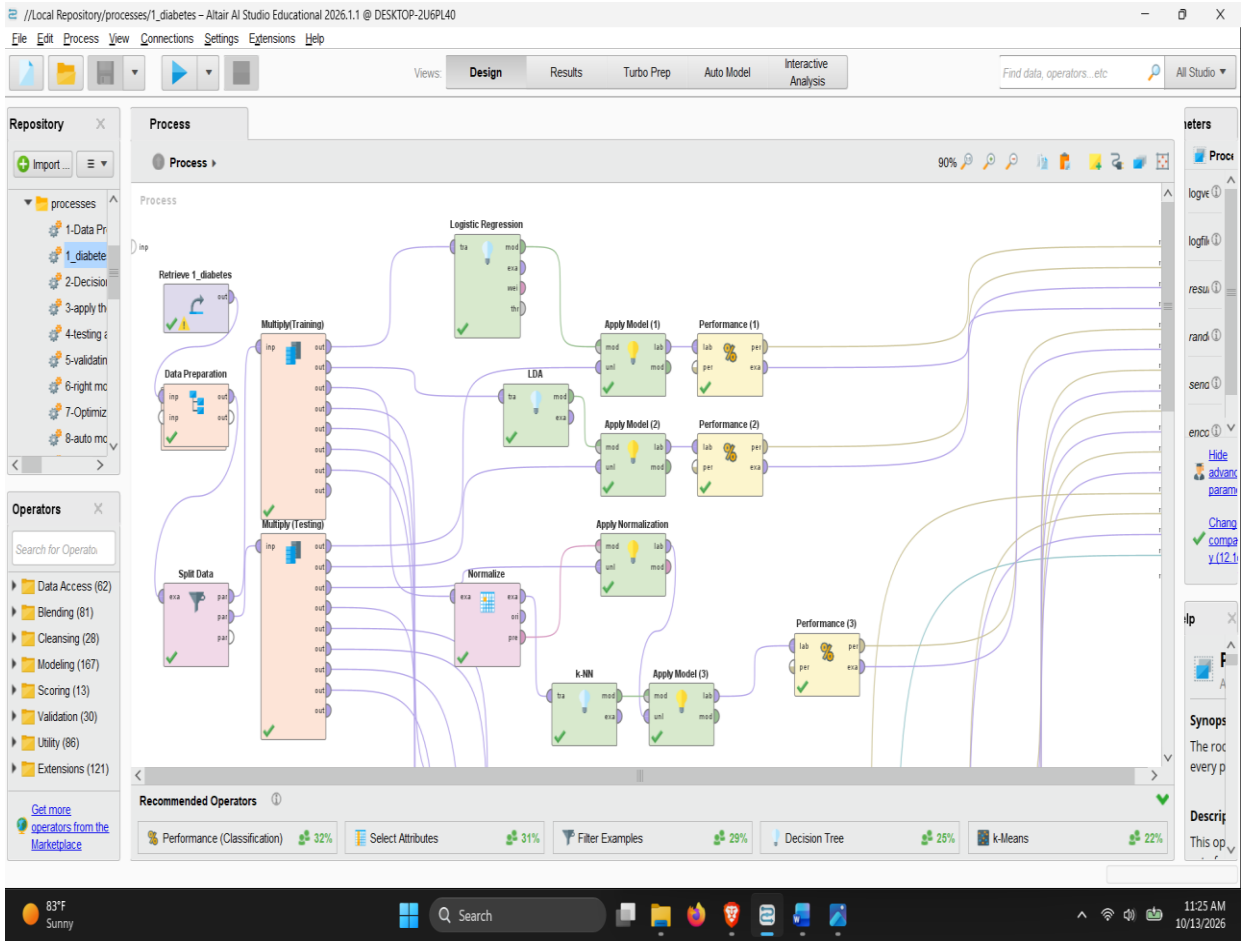
Six classification approaches were compared on the same 230-record holdout set

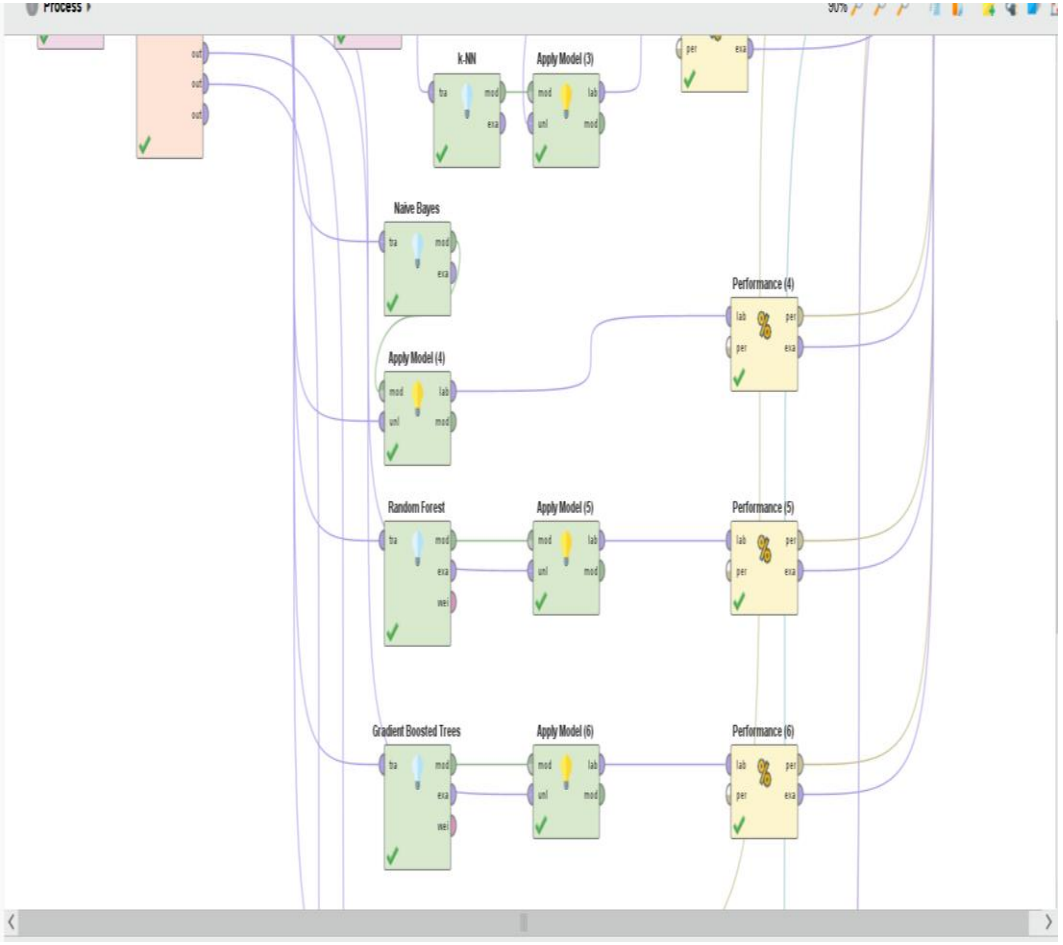
- **Logistic Regression** (primary interpretable binary classification model.
- **Linear Discriminant Analysis** (Linear decision rule for class separation)
- **K- Nearest Neighbors** (Distance-based nonparametric classifier, normalized inputs)
- **Naïve Bayes** (Probability-based baseline classifier)
- **Random Forest** (Bagging ensemble of decision trees)
- **Gradient Boosted Trees** (Sequential boosting model that corrects prior errors)

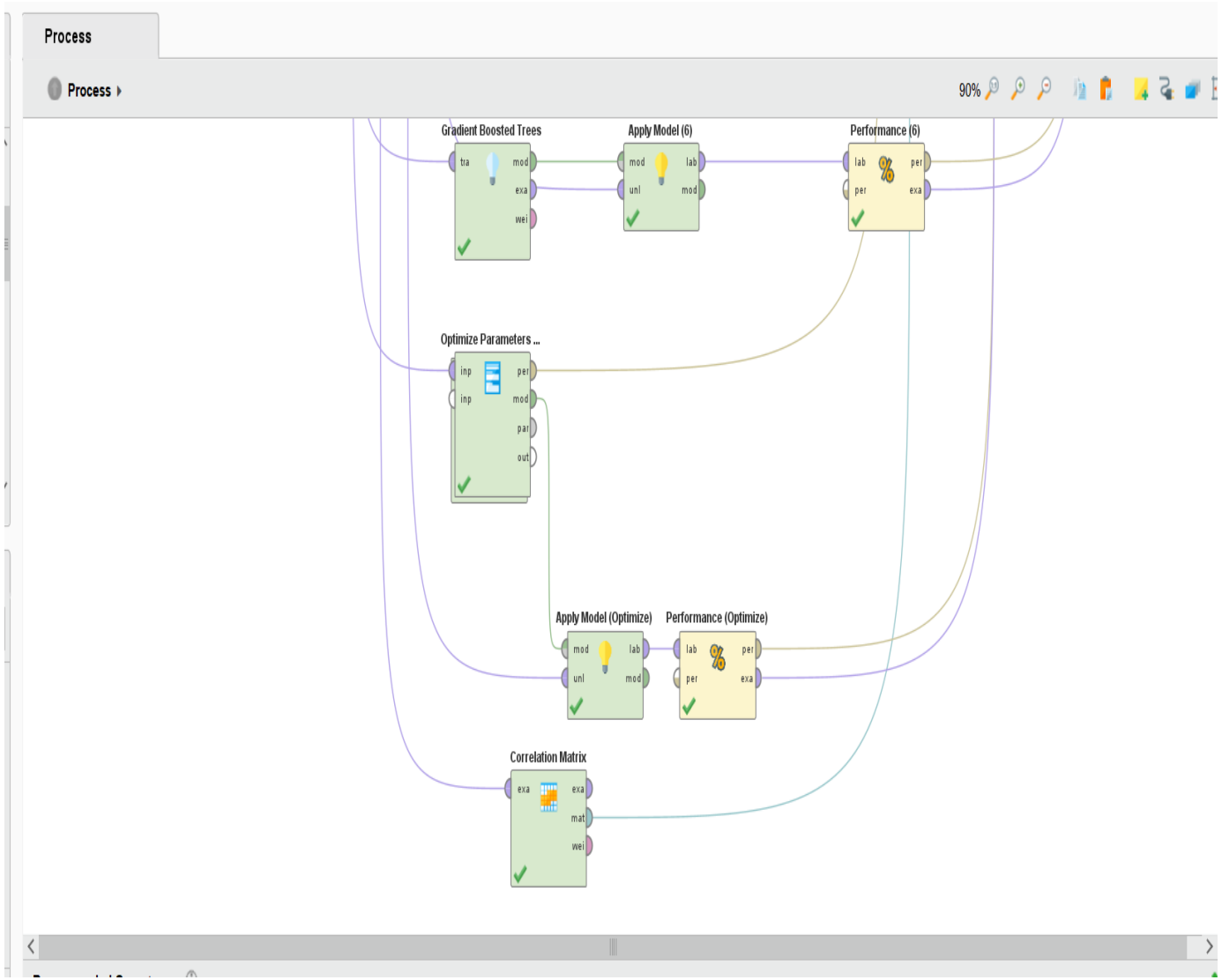
Common evaluation: Accuracy • AUC (when available) • Precision • Recall/Sensitivity • F1 • Specificity • Confusion Matrix

7. Modeling Workflow in Altair AI Studio

A single process was used to prepare, train, apply, evaluate, optimize, and compare models







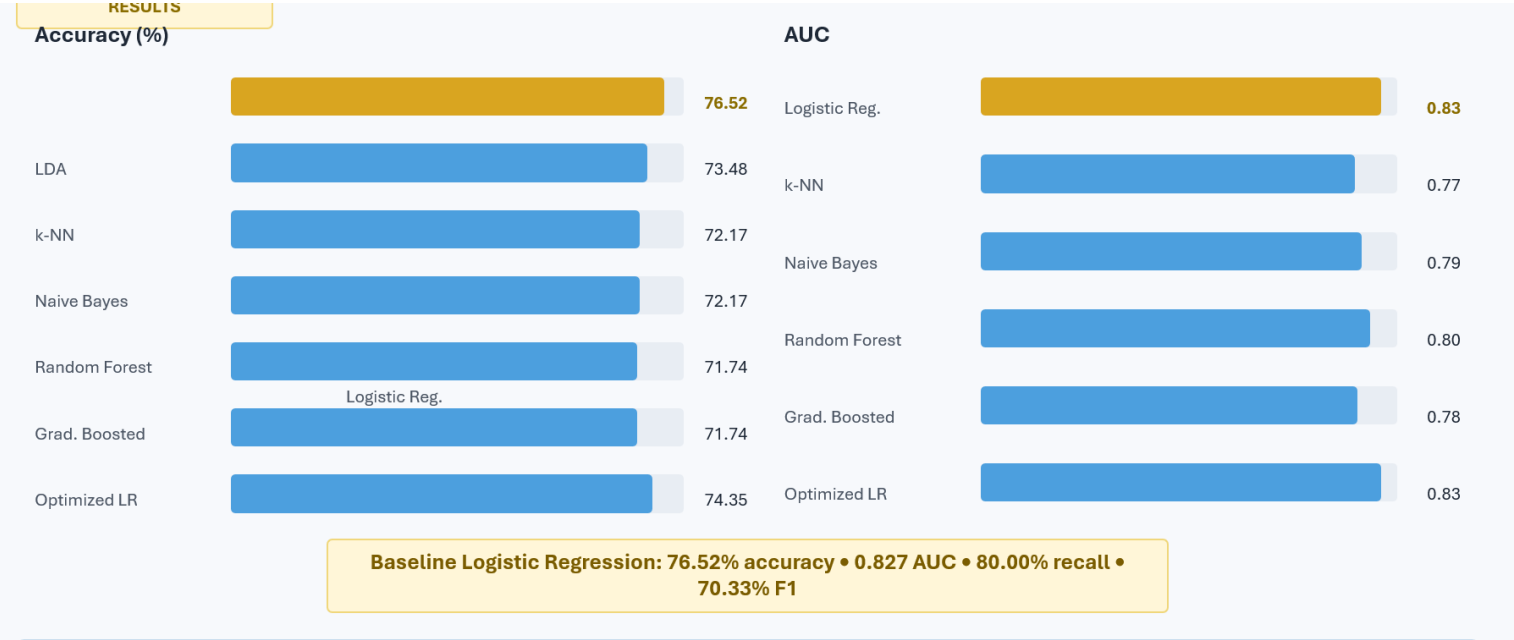
Experimental design

- 70% training / 30% holdout testing
- Same holdout used for model comparison
- Performance operators returned confusion matrices and metrics
- Logistic Regression tuning used nested 10-fold cross-validation
- Correlation analysis remained descriptive and did not affect training

The holdout set remained separate from hyperparameter selection, supporting a fair final comparison.

8. Holdout Performance Comparison

Baseline Logistic Regression achieved the strongest overall balance across key metrics



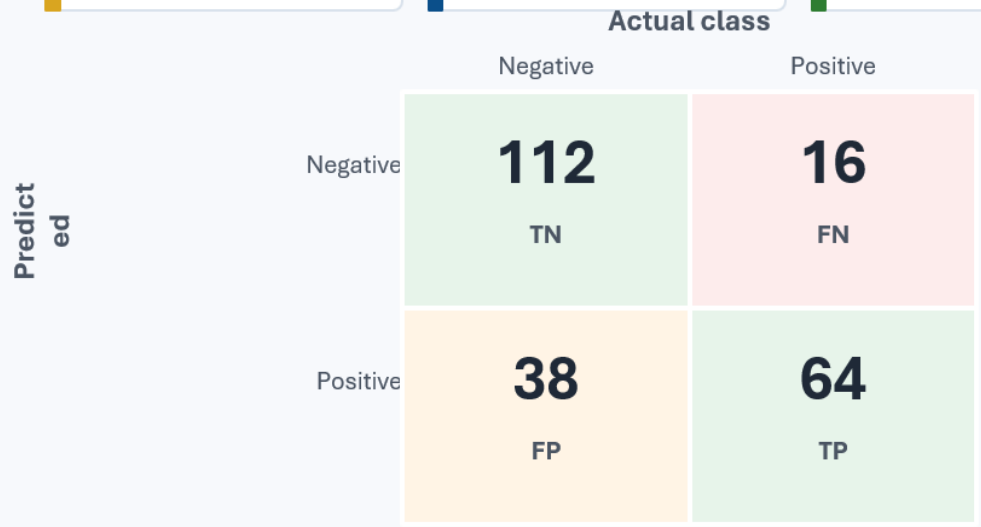
Baseline Logistic Regression: 76.52% accuracy • 0.827 AUC • 80.00% recall • 70.33% F1

Random Forest had the highest specificity (90.67%) but only 36.25% recall, missing many positive cases.

9. Baseline Logistic Regression Performance

Recommended model before tuning because it detected positive cases well while maintaining solid specificity

76.52%	0.827	80.00%	70.33%	74.67%
Accuracy	AUC	Recall	F1 score	Specificity



How to interpret the errors

- 112 of 150 negative cases were correctly classified.
- 64 of 80 positive cases were correctly identified.
- Only 16 positive cases were missed (false negatives).
- AUC = 0.827 indicates strong ranking ability across thresholds.

For screening-style use, the relatively low false-negative count supports choosing the baseline Logistic Regression mod

10. Logistic Regression Optimization

Grid search improved specificity, but holdout recall and F1 decreased

- Grid-search setup:
- Optimize Parameters (Grid)
- 10-fold cross-validation
- Lambda: 0.001 to 1
- Logarithmic scale, 11 values
- Best CV accuracy ≈ 77.2%
- CV performance: 77.16% ± 4.45%

Baseline vs. optimized holdout performance

Metric	Baseline	Optimized	Change
Accuracy	76.52%	74.35%	−2.17 pts
AUC	0.827	0.827	No change
Recall	80.00%	66.25%	−13.75 pts
F1	70.33%	64.24%	−6.09 pts
Specificity	74.67%	78.67%	+4.00 pts

Decision: keep baseline Logistic Regression

A better cross-validation setting did not translate into better holdout detection—final evaluation on unseen data rema

11. Key Insights, Limitations and Conclusion

What the project demonstrates and how the final model should be interpreted

Key insights

- Model complexity did not guarantee better generalization.
- Glucose and BMI showed clear differences between Outcome groups.
- Predictive separation is multivariable rather than driven by one feature.
- False-negative cost matters when selecting a screening-oriented model.

Limitations

- Small benchmark dataset
- Population limited to adult women of Pima heritage
- Large invalid-zero counts for Insulin and SkinThickness
- Mean imputation may influence results
- External validation is required before clinical use

Final recommendation

Baseline Logistic Regression

76.52% accuracy
0.827 AUC
80.00% recall
70.33% F1

Use as an educational predictive model—not as a medical diagnosis system.

The baseline Logistic Regression model provided the best overall balance and was retained as the final model.